

Supplementary Material

S1. Cohort Flow and Missingness

Of 267 subjects in the spine-generic dataset (release r20231212), 203 were healthy controls (HC) and 64 had pathology (62 mild cervical compression; 2 degenerative cervical myelopathy). Both cohorts were pre-specified prior to analysis.

Missingness patterns. Among HC subjects, 15 (7.4%) were missing one or more biomarkers: MTR ($n = 13$), MTsat ($n = 18$), FA/MD/AD/RD ($n = 7$ each). Missingness was attributable to acquisition failure or automated QC rejection during pipeline processing (SCT v7.4 QC metrics including PAM50 template overlap, diffusion tensor model-fit residuals, and MT saturation map artifact screening), not to demographic or vendor characteristics. Among ALL subjects, 21 (7.9%) were incomplete. The complete-case HC cohort comprised $N = 188$ and the complete-case ALL cohort $N = 246$.

Table S1. Demographic characteristics by vendor (ALL cohort, $N = 246$).

Characteristic	GE ($n = 33$)	Philips ($n = 47$)	Siemens ($n = 166$)	Total ($n = 246$)	p
Sites (n)	6	8	30	44	—
Age (years)	28.0 ± 5.2	29.4 ± 4.7	30.0 ± 6.3	29.6 ± 5.9	0.193
Sex (% female)	48.5	46.8	47.0	47.2	0.986
Pathology (HC/MC/DCM)	25/8/0	33/13/1	130/35/1	188/56/2	0.661

S2. Biomarker Extraction and Quality Control

All biomarkers were extracted using the Spinal Cord Toolbox (SCT v7.4; <https://spinalcordtoolbox.com>) following the spine-generic processing pipeline documentation (<https://spine-generic.readthedocs.io>).

Processing steps:

- **C2–C3 level identification:** Automated PAM50 template registration.
- **CSA:** Deep-learning segmentation (`sct_deepseg_sc`) on T2w images.
- **GM-CSA:** Gray matter segmentation via `sct_deepseg_gm` (T2*-weighted contrast).
- **MTR/MTsat:** Computed from MT-on and MT-off acquisitions.
- **DTI metrics:** Diffusion tensor fitting via `sct_dmri_compute_dti`.

QC rejection criteria: Segmentation overlap with PAM50 template < 0.6 ; DTI model-fit residual > 2 SD above cohort mean; MT saturation map visual artifact flag.

S3. Harmonization Methods: Mathematical Details

S3.1 ComBat (per-metric). For each biomarker y_{ijv} for subject i at site v :

$$y_{ijv}^{\text{harm}} = \frac{y_{ijv} - \hat{\alpha}_v - X_{ij}\hat{\beta}}{\hat{\gamma}_v} + X_{ij}\hat{\beta}$$

where $\hat{\alpha}_v$ and $\hat{\gamma}_v$ are site-specific location and scale parameters estimated with empirical Bayes shrinkage, and X_{ij} is the design matrix of biological covariates (age, sex). This is the standard ComBat formulation [8,9] applied independently to each biomarker.

S3.2 ComBat-joint. All eight biomarkers are modeled jointly with a shared site parameter estimated via joint empirical Bayes (`eb=True` in `neuroCombat`). The mathematical form is identical to S3.1 but $\hat{\alpha}_v$ and $\hat{\gamma}_v$ are estimated simultaneously across biomarkers, preserving cross-metric covariance structure.

S3.3 CovBat. After ComBat-joint harmonization, the residual matrix E (subjects $\times$ biomarkers) is decomposed via PCA: $E = ZW^T$, where Z is the PC score matrix and W is the loading matrix. PCs explaining at least 95% of the residual variance were retained ($K = 5$ of 8 in both cohorts; see Supplementary Fig. S4 for the resulting covariance structure). Each PC k is then harmonized independently: $Z_k^{\text{harm}} = (Z_k - \hat{\alpha}_{vk})/\hat{\gamma}_{vk}$, where $\hat{\alpha}_{vk}$ and $\hat{\gamma}_{vk}$ are site-specific parameters for PC k . The harmonized data are reconstructed as $Y^{\text{harm}} = Y^{\text{ComBat-joint}} + Z^{\text{harm}}W^T$.

S3.4 RELIEF. After ComBat-joint harmonization, residuals undergo ICA decomposition: $E = SA^T$, where S is the independent component score matrix and A is the mixing matrix. For each component c , a one-way ANOVA tests vendor association. Components with Benjamini-Hochberg FDR-controlled $q < 0.05$ are zeroed ($S_c \leftarrow 0$). The remaining components are recombined: $E^{\text{RELIEF}} = S^{\text{cleaned}}A^T$.

S3.5 LME/FE. For each biomarker:

$$y_{ijv} = \beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Sex}_i + u_v + \varepsilon_{ijv}$$

where $u_v \sim \mathcal{N}(0, \sigma_u^2)$ is the vendor-specific random intercept. Harmonized values: $y_{ijv}^{\text{harm}} = y_{ijv} - \hat{u}_v$ when LME converges; $y_{ijv}^{\text{harm}} = y_{ijv} - \bar{y}_v + \bar{y}$ (OLS centering) when the random-effects covariance matrix is singular (FE fallback).

S4. Evaluation Metrics: Computational Details

S4.1 Vendor η^2 . One-way ANOVA: $\eta^2 = \text{SS}_{\text{vendor}}/\text{SS}_{\text{total}}$. Reported as mean across 8 biomarkers.

S4.2 PERMANOVA. Euclidean distance matrix on the 8-dimensional biomarker space after per-dimension z-scoring. 9,999 permutations of vendor labels. $R^2 = \text{SS}_{\text{vendor}}/\text{SS}_{\text{total}}$ from the distance-based pseudo- F statistic [20]. Implementation: custom NumPy code

(pseudo- F and permutation loop written in-house; no third-party PERMANOVA package). On z-scored features with Euclidean distance, the multivariate R^2 equals the mean of the per-dimension η^2 computed on the same subjects. Because all six methods are fitted and evaluated on the same unified complete-case sample (HC $N = 188$, ALL $N = 246$), this identity holds exactly for every method, so univariate and multivariate summaries are directly comparable.

S4.3 Biological R^2 . Type-II semi-partial R^2 from linear models: biomarker $\sim$ Age + Sex. Reported as mean across biomarkers.

S4.4 Subject-level preservation. Pearson correlation between pre- and post-harmonization 8-dimensional biomarker vectors, computed per subject and averaged:

$$r = \text{mean}_i[\text{corr}(y_i^{\text{pre}}, y_i^{\text{post}})].$$

S4.5 Unified complete-case analysis sample. All results in the main text and this supplement are computed on a single pre-specified complete-case sample per cohort: HC $N = 188$ and ALL $N = 246$ (all eight biomarkers plus age, sex, and vendor observed). ComBat and LME/FE were refitted on this unified sample (the ComBat and LME/FE columns of every table therefore derive from these refits); ComBat-joint, CovBat, and RELIEF are complete-case by construction because they require the full joint biomarker panel. This unified definition replaces the earlier per-metric maximum-available-sample evaluation and removes any sample mismatch between univariate and multivariate endpoints.

S4.6 Within-resample paired bootstrap. To compare $r_{\text{pre,post}}$ between method pairs, each stratified bootstrap resample ($B = 2,000$, resampling subjects within vendor strata) was passed to *all* methods simultaneously; the per-resample differences $r_A - r_B$ form the null distribution for the pairwise test. This pairing removes between-resample variance (the source of the overlapping marginal CIs) and directly targets the pairwise contrast. Results: in HC, 8 of 10 method pairs were significantly separated (all $p \leq 0.003$; ComBat vs. CovBat $p = 0.070$ and ComBat vs. RELIEF $p = 0.694$ were not); in ALL, 9 of 10 pairs were separated (ComBat vs. RELIEF $p = 0.403$ was not). LME/FE was significantly highest and CovBat significantly lowest in both cohorts. Full results in Supplementary Table S7.

Table S7. Within-resample paired bootstrap comparison of subject-level preservation ($r_{\text{pre,post}}$) between all method pairs ($B = 2,000$, stratified by vendor; complete-case sample). $\Delta r = r_A - r_B$; CI = 95% percentile interval of the per-resample differences; p = two-sided bootstrap p-value. HC cohort above, ALL cohort below.

Cohort	Comparison	Δr	95% CI	p	Significant
HC	ComBat vs. ComBat-joint	-0.017	[-0.020, -0.014]	0.0005	Yes
	ComBat vs. CovBat	0.007	[-0.000, 0.015]	0.070	No
	ComBat vs. RELIEF	-0.001	[-0.007, 0.005]	0.694	No
	ComBat vs. LME/FE	-0.027	[-0.033, -0.020]	0.0005	Yes
	ComBat-joint vs. CovBat	0.024	[0.017, 0.032]	0.0005	Yes
	ComBat-joint vs. RELIEF	0.016	[0.010, 0.022]	0.0005	Yes
	ComBat-joint vs. LME/FE	-0.010	[-0.015, -0.004]	0.003	Yes
	CovBat vs. RELIEF	-0.009	[-0.015, -0.003]	0.002	Yes
	CovBat vs. LME/FE	-0.034	[-0.043, -0.025]	0.0005	Yes
	RELIEF vs. LME/FE	-0.026	[-0.033, -0.018]	0.0005	Yes
ALL	ComBat vs. ComBat-joint	-0.013	[-0.015, -0.011]	0.0005	Yes
	ComBat vs. CovBat	0.011	[0.005, 0.018]	0.001	Yes
	ComBat vs. RELIEF	0.002	[-0.003, 0.008]	0.403	No
	ComBat vs. LME/FE	-0.027	[-0.033, -0.020]	0.0005	Yes
	ComBat-joint vs. CovBat	0.024	[0.018, 0.031]	0.0005	Yes
	ComBat-joint vs. RELIEF	0.015	[0.010, 0.020]	0.0005	Yes
	ComBat-joint vs. LME/FE	-0.014	[-0.019, -0.008]	0.0005	Yes
	CovBat vs. RELIEF	-0.009	[-0.015, -0.004]	0.001	Yes
	CovBat vs. LME/FE	-0.038	[-0.046, -0.029]	0.0005	Yes
	RELIEF vs. LME/FE	-0.029	[-0.035, -0.022]	0.0005	Yes

S5. LOVO External Validation: Protocol

Training. For each method, harmonization models were fitted on subjects from two vendors (training set). The third vendor was entirely held out as test set.

Transformation of unseen vendors:

- **ComBat/ComBat-joint:** Test-vendor biomarker values were z-scored to the training-vendor harmonized distribution: $y_{\text{test}}^{\text{harm}} = (y_{\text{test}} - \mu_{\text{test}}) \times (\sigma_{\text{train}}^{\text{harm}} / \sigma_{\text{test}}) + \mu_{\text{train}}^{\text{harm}}$.
- **CovBat:** Test-vendor residuals were projected onto training PCA space; PC scores were z-score matched to training PC-score distribution.
- **RELIEF:** Test-vendor residuals were projected onto training ICA space via $\text{inv}(A^T)$; batch-flagged components were zeroed using the training-vendor flag set.
- **LME/FE:** Test-vendor offset set to 0; test-vendor values were carried through essentially untransformed (raw values, no vendor or covariate adjustment).

Evaluation. The same three endpoints were computed after pooling the harmonised training-vendor and held-out test-vendor data within each fold, then averaged across the 3 rotation folds. Because the transformation applied to the test vendor is per-biomarker

affine (location–scale) for ComBat, ComBat-joint, and LME/FE, the test-vendor pre/post correlation equals 1.0 by construction for these methods; their preservation statistics therefore reflect the pooled sample rather than the unseen vendor in isolation. For CovBat and RELIEF, projection onto the training PCA/ICA basis mixes biomarker dimensions, and the test-vendor pre/post correlation fell below 1 ($r = 0.85$ – 0.97 across folds), providing a genuine held-out-vendor distortion measure (Table S6).

Table S6. Test-vendor-only evaluation metrics by held-out vendor and method (mean across 8 biomarkers; complete-case training sample $N = 188$). |shift|: mean absolute shift of the held-out-vendor mean relative to the pooled training distribution, in training-SD units; var ratio: held-out-vendor to training variance ratio; centroid shift: multivariate centroid shift in pooled SD units; $r_{\text{pre,post}}$: test-vendor pre/post correlation. ComBat and ComBat-joint force the test-vendor distribution onto the training template (shift exactly 0 by construction); CovBat leaves small residual shifts; RELIEF and LME/FE leave the held-out vendor shifted relative to training.

Held-out	Method	shift	var ratio	centroid shift	$r_{\text{pre,post}}$ (test)
GE	Original	1.46	0.98	5.68	1.000
	ComBat	0.00	1.00	0.00	1.000
	ComBat-joint	0.00	1.00	0.00	1.000
	CovBat	0.03	0.94	0.12	0.851
	RELIEF	1.64	1.12	5.97	0.925
	LME/FE	1.62	1.08	5.92	1.000
Philips	Original	0.80	0.83	2.58	1.000
	ComBat	0.00	1.00	0.00	1.000
	ComBat-joint	0.00	1.00	0.00	1.000
	CovBat	0.03	1.00	0.10	0.949
	RELIEF	0.85	0.97	2.69	0.973
	LME/FE	0.92	0.99	3.14	1.000
Siemens	Original	1.05	0.93	3.37	1.000
	ComBat	0.00	1.00	0.00	1.000
	ComBat-joint	0.00	1.00	0.00	1.000
	CovBat	0.05	0.96	0.16	0.911
	RELIEF	1.42	1.24	4.54	0.960
	LME/FE	1.39	1.13	4.51	1.000

S6. Design C Balanced-Subsample Analysis

To isolate the effect of vendor imbalance, 10 subjects were randomly drawn from each vendor (HC cohort only), yielding 30-subject balanced subsets. Five-fold internal validation was performed per subset. This procedure was repeated for 1,000 bootstrap iterations. Results are shown in Supplementary Fig. S2.

S7. Supplementary Figures

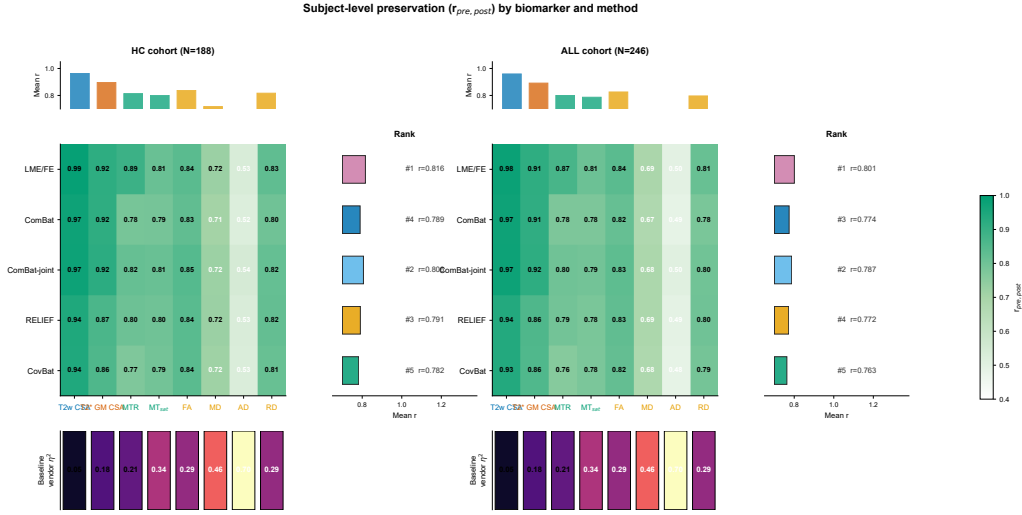


Figure 1: Subject-level preservation ($r_{pre,post}$) by method. Marginal distributions of per-subject Pearson correlations between pre- and post-harmonization 8-dimensional biomarker vectors. Higher values indicate better preservation of subject ranking. LME/FE shows highest preservation ($r = 0.82$); CovBat lowest ($r = 0.78$).

Design C: Robustness to balanced vendor composition (20 random seeds, $n \approx 98/\text{seed}$)

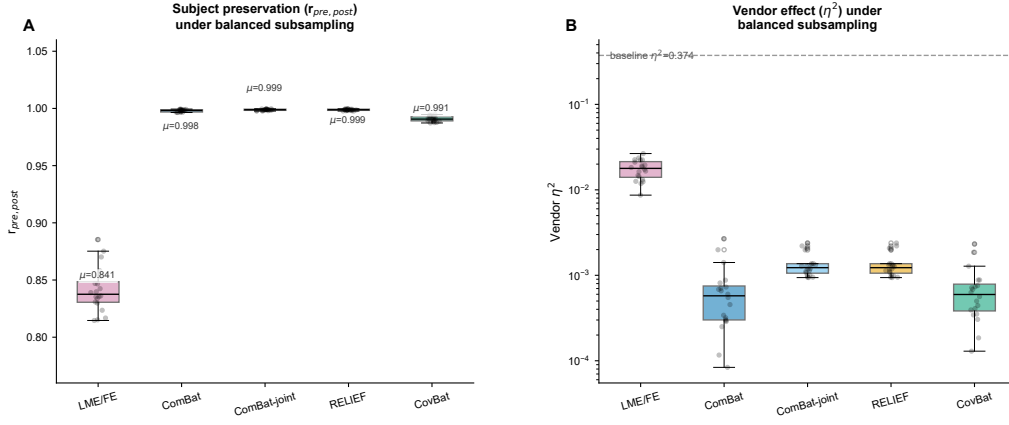


Figure 2: Design C balanced-subsample robustness analysis. (a) $r_{\text{pre,post}}$ distributions across 1,000 bootstrap samples (10 subjects per vendor). (b) PERMANOVA R^2 distributions. Under balanced conditions the preservation hierarchy reversed: ComBat-family methods converged to near-perfect preservation ($r = 0.99$ – 1.00 ; CovBat 0.991 , others ≥ 0.998) while LME/FE remained at $r = 0.841$. The main-analysis preservation gradient therefore largely reflects real-world vendor imbalance (Siemens 67%) rather than intrinsic method limitations.

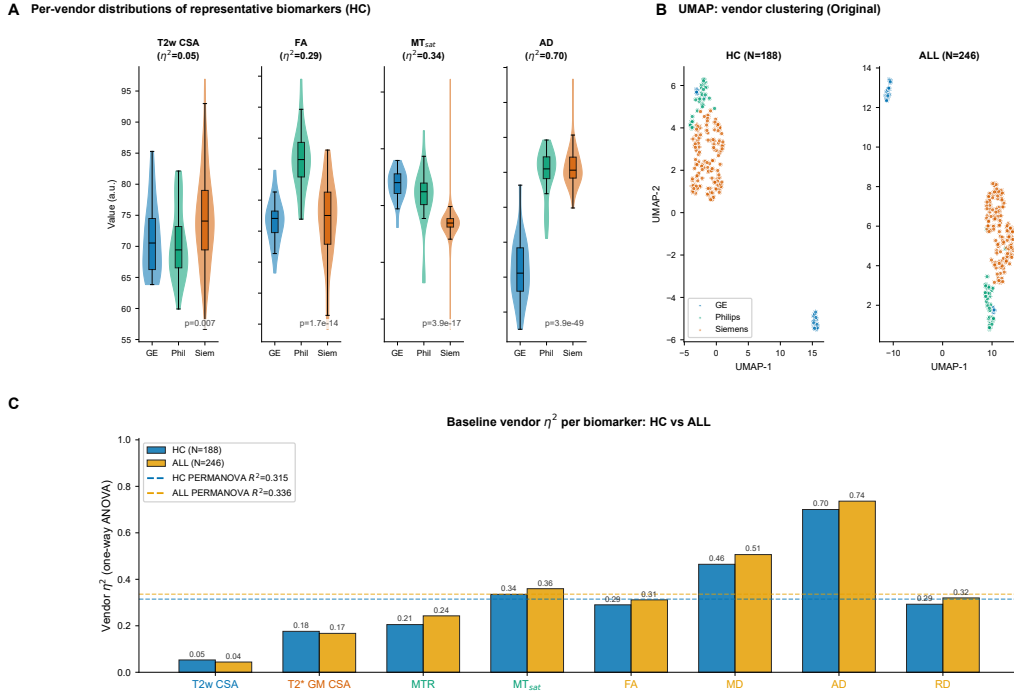


Figure 3: Baseline vendor effects in spinal cord qMRI biomarkers. **(a)** Per-vendor distributions of four representative biomarkers in the HC cohort (T2w-CSA $\eta^2 = 0.05$; FA 0.29; MTsat 0.34; AD 0.70; one-way ANOVA p values shown). **(b)** UMAP embedding of the 8-dimensional biomarker space (Original, unharmonised data) coloured by vendor, for the HC ($N = 188$) and ALL ($N = 246$) cohorts. **(c)** Per-biomarker baseline vendor η^2 in the HC and ALL cohorts; mean $\eta^2 = 0.31$ (HC) and 0.34 (ALL); PERMANOVA $R^2 = 0.315$ (HC) and 0.336 (ALL), both $p = 1 \times 10^{-4}$ (permutation lower bound with 9,999 permutations).

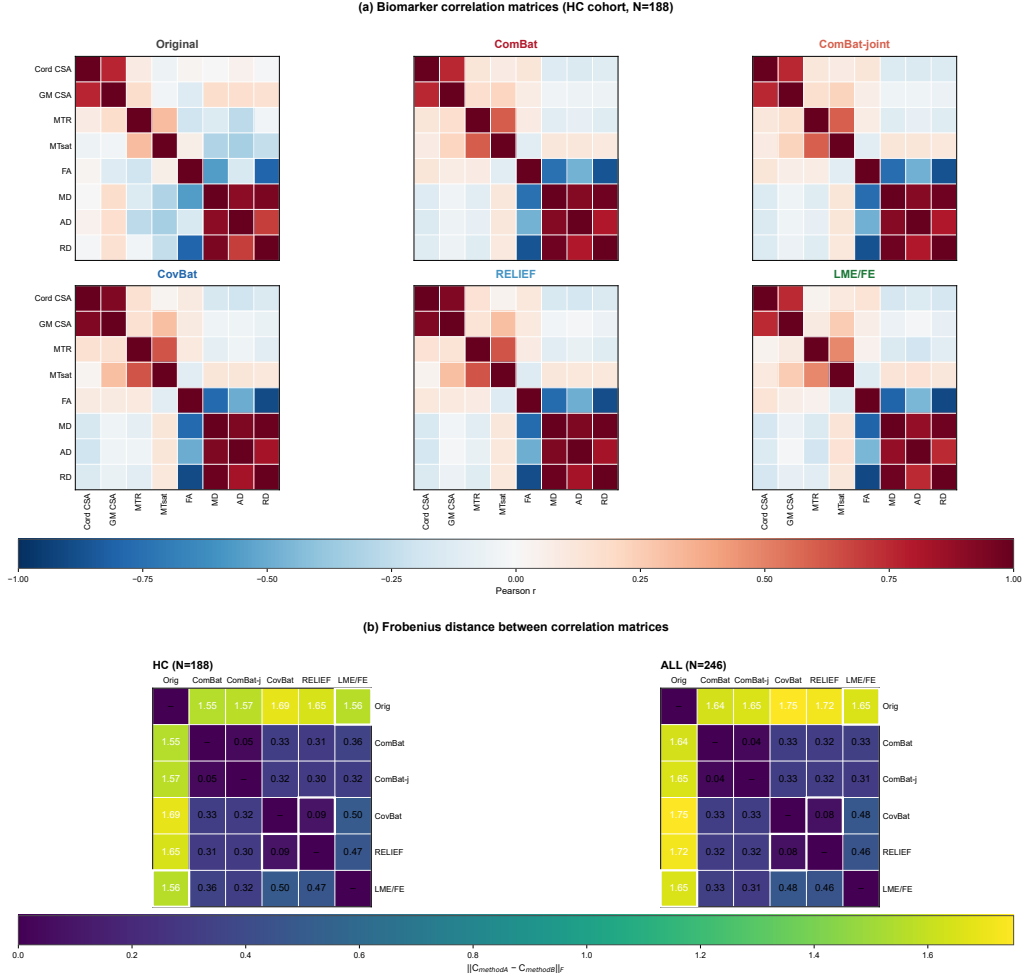


Figure 4: Cross-metric covariance structure preservation. **(a)** Pearson correlation matrices of the eight biomarkers (HC cohort, $N = 188$) before and after each method. **(b)** Frobenius distance between the correlation matrices of each pair of methods (lower = more similar covariance structure). CovBat and RELIEF are the closest (HC: 0.09, ALL: 0.08), indicating near-identical covariance structure despite the additional ICA round-trip in RELIEF. ComBat-joint and ComBat are also close (HC: 0.05, ALL: 0.04), showing that joint empirical Bayes has limited impact on the covariance structure beyond the per-metric ComBat baseline. ComBat (HC: 1.55, ALL: 1.64) and LME/FE (HC: 1.56, ALL: 1.65) remain closest to the original, reflecting minimal covariance distortion.

S8. Supplementary Data

The following data files are provided:

- `biomarkers_master.csv`: Raw biomarker values with demographics (267 subjects)
- `demographics_completeness.csv`: Complete-case analysis sample demographics by vendor and cohort (Supplementary Table S1)
- `lovo_summary.csv`: Leave-one-vendor-out validation results (18 rows)
- `lovo_testvendor.csv`: Leave-one-vendor-out held-out (test) vendor shifts and cor-

relations per fold, method, and biomarker (162 rows; Supplementary Table S6)

- `corr_matrices_HC.csv` / `corr_matrices_ALL.csv`: Full 8×8 Pearson correlation matrices for all six methods (Supplementary Fig. S4)
- `frobenius_distance_HC.csv` / `frobenius_distance_ALL.csv`: Pairwise Frobenius distances between method correlation matrices (Supplementary Fig. S4)
- `comparison_long.csv`: Per-biomarker evaluation metrics (vendor η^2 , age R^2 , sex R^2 , $r_{\text{pre,post}}$) for all methods and both cohorts on the unified complete-case sample (96 rows)
- `cc_summary.csv`: Method-level summary (mean/median/min/max) of the per-biomarker metrics in `comparison_long.csv`
- `permanova_results.csv`: PERMANOVA results (HC and ALL cohorts)
- `bootstrap_r_ci_cc.csv` / `bootstrap_eta2_baseline_cc.csv`: Bootstrap 95% confidence intervals for $r_{\text{pre,post}}$ and baseline η^2 ($B = 2,000$, stratified by vendor)
- `bootstrap_paired_r.csv`: Paired bootstrap tests of $r_{\text{pre,post}}$ differences between method pairs (20 rows; Supplementary Table S7)
- `lme_effect_summary.csv`: LME convergence status per biomarker $\times$ cohort
- `relief_components_HC.csv` / `relief_components_ALL.csv`: RELIEF ICA component statistics

The analysis code is available at <https://github.com/wudengke2010/spine-generic-harmonization>.